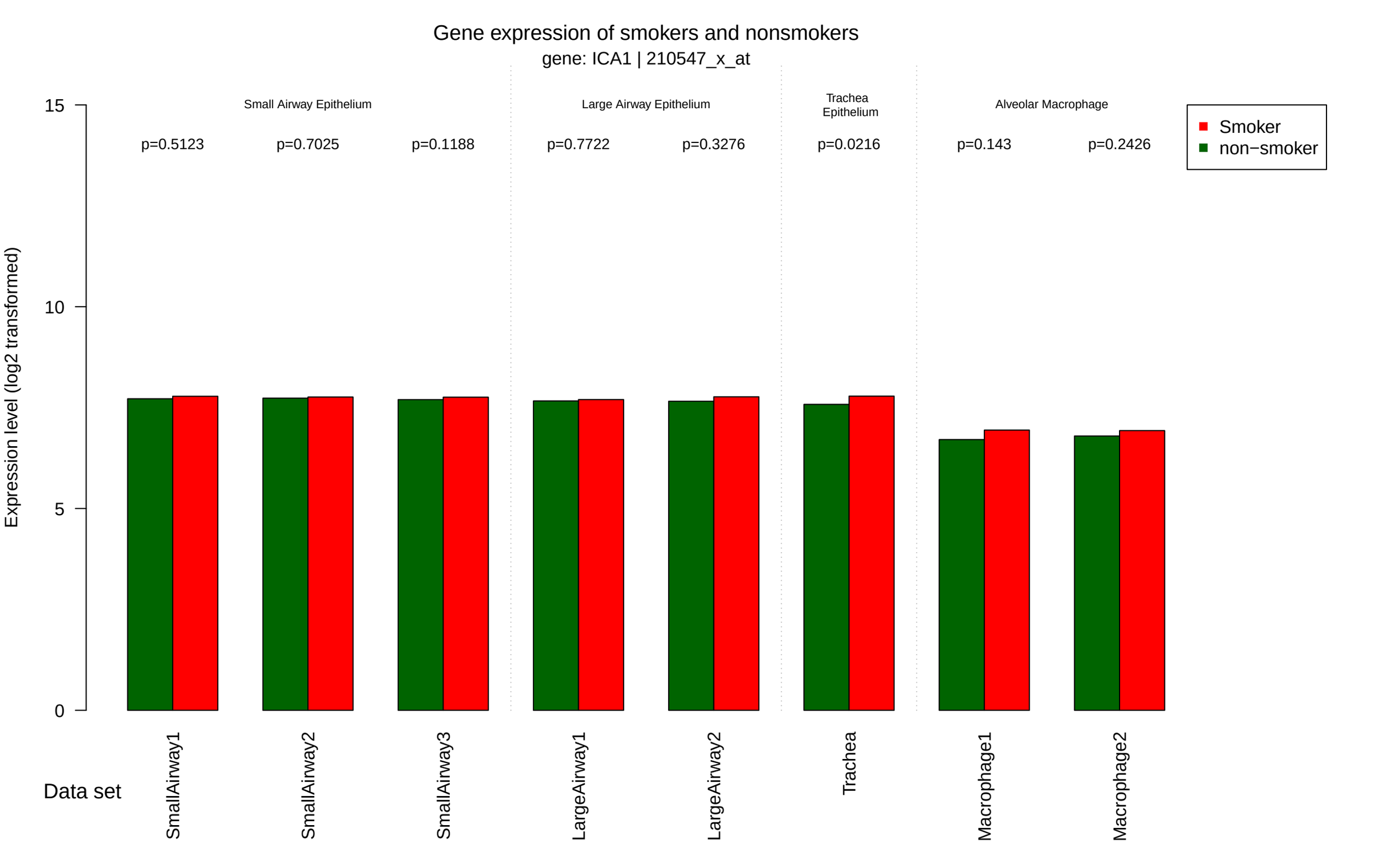
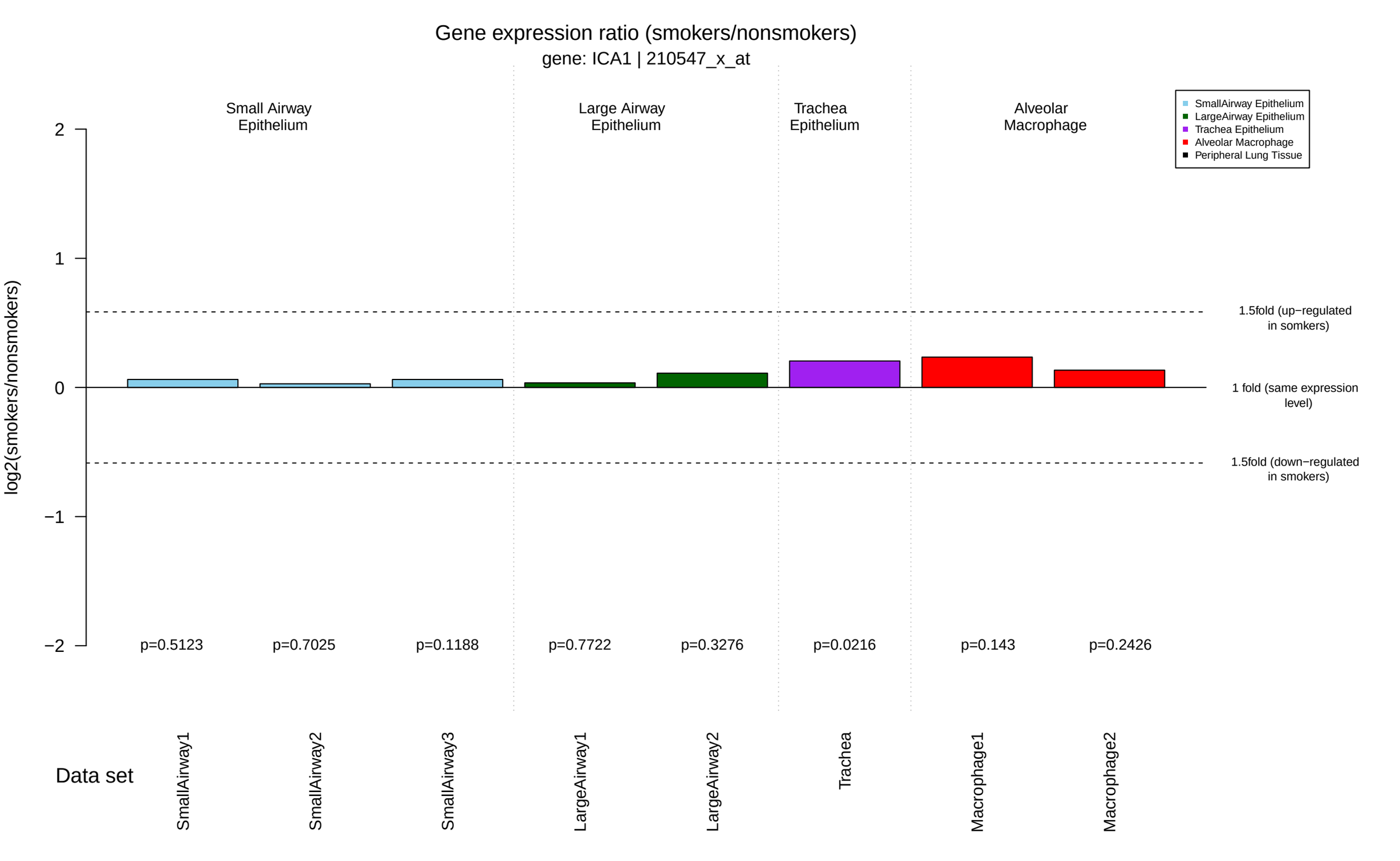
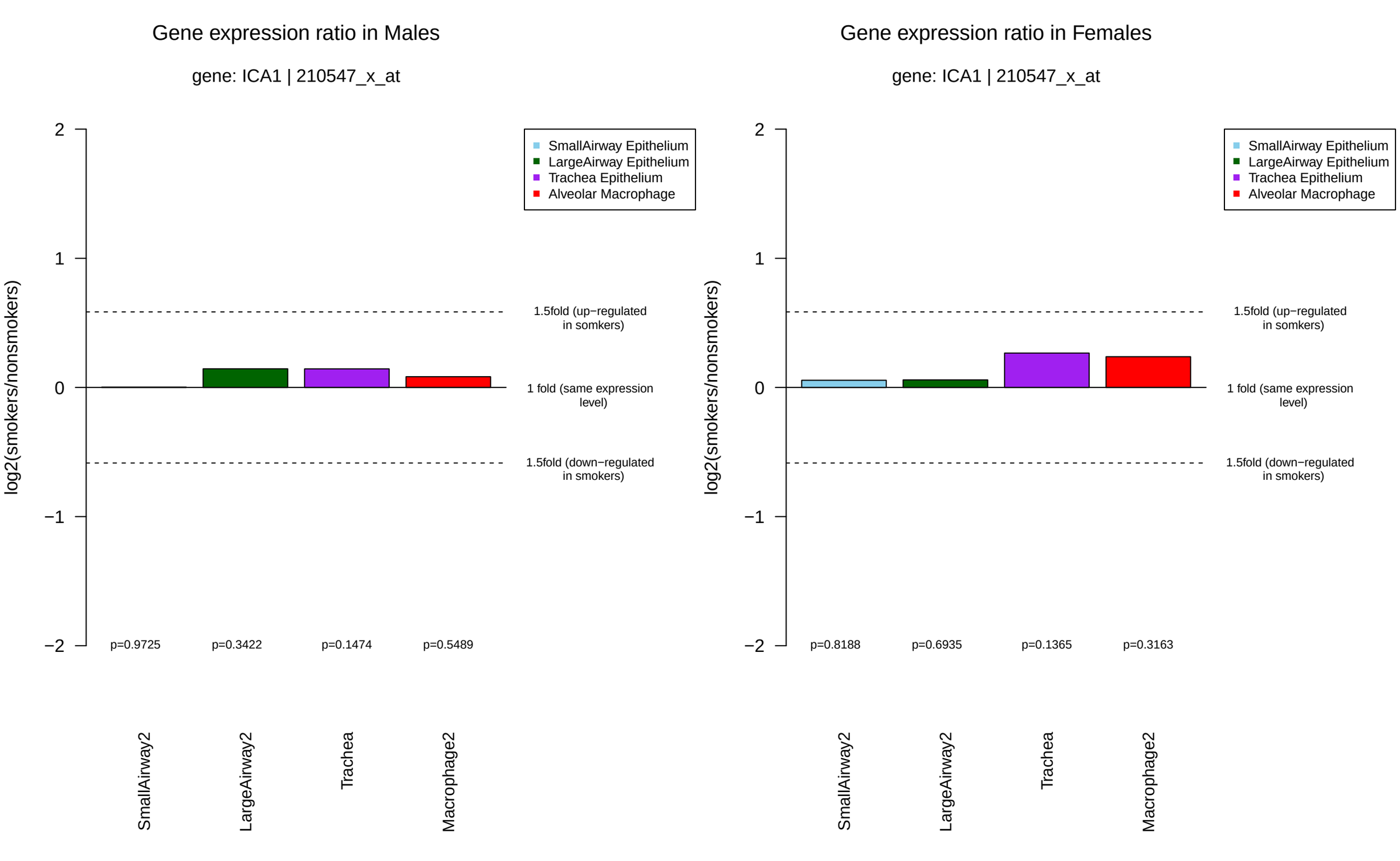
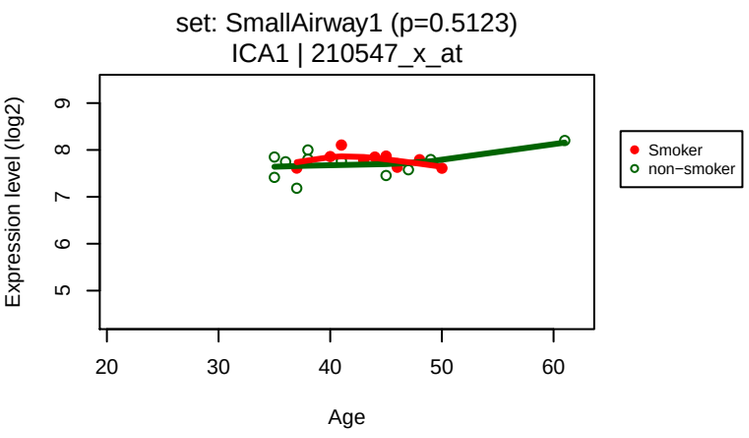
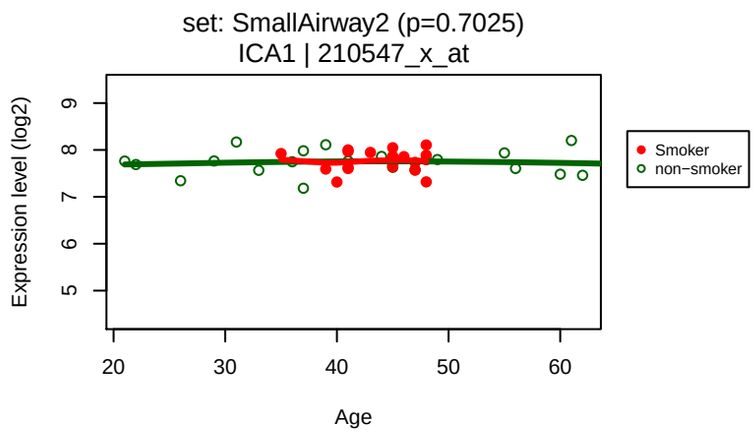
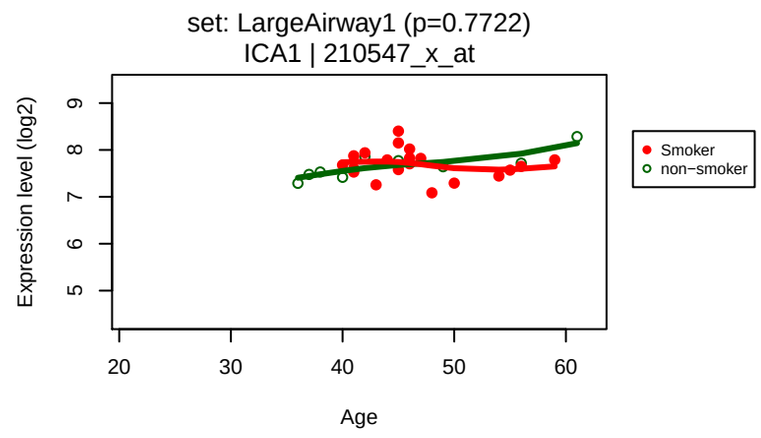

Gene expression of smokers and nonsmokers
gene: ICA1 | 210547_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5123
p=0.7025
p=0.1188
p=0.7722
p=0.3276
p=0.0216
p=0.143
p=0.2426
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

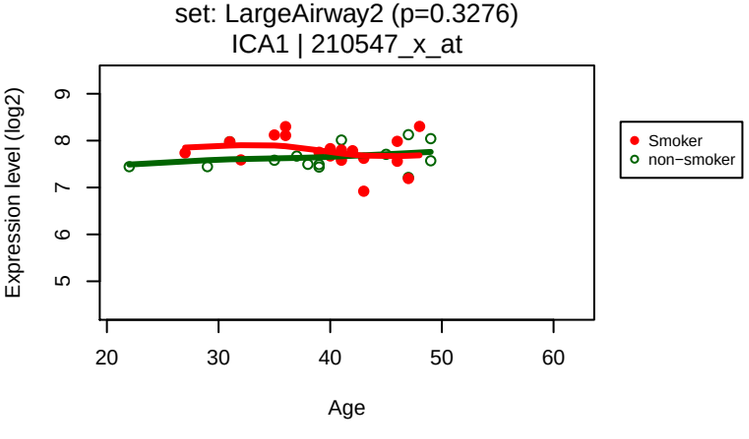

Gene expression ratio (smokers/nonsmokers)
gene: ICA1 | 210547_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5123
p=0.7025
p=0.1188
p=0.7722
p=0.3276
p=0.0216
p=0.143
p=0.2426
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

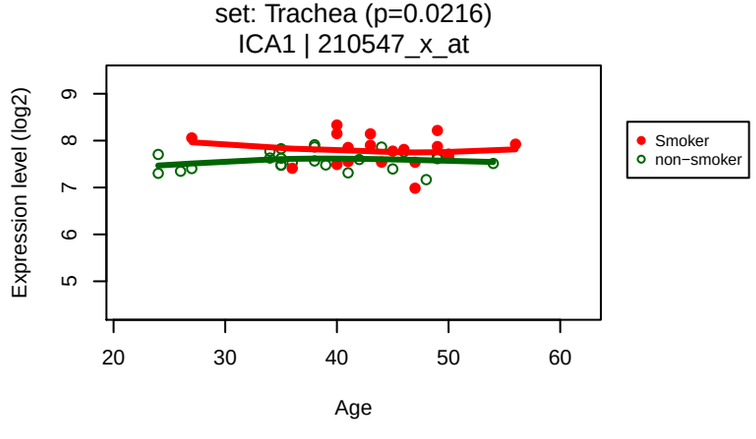

Gene expression ratio in Males
gene: ICA1 | 210547_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9725
p=0.3422
p=0.1474
p=0.5489
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ICA1 | 210547_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8188
p=0.6935
p=0.1365
p=0.3163
SmallAirway2
LargeAirway2
Trachea
Macrophage2

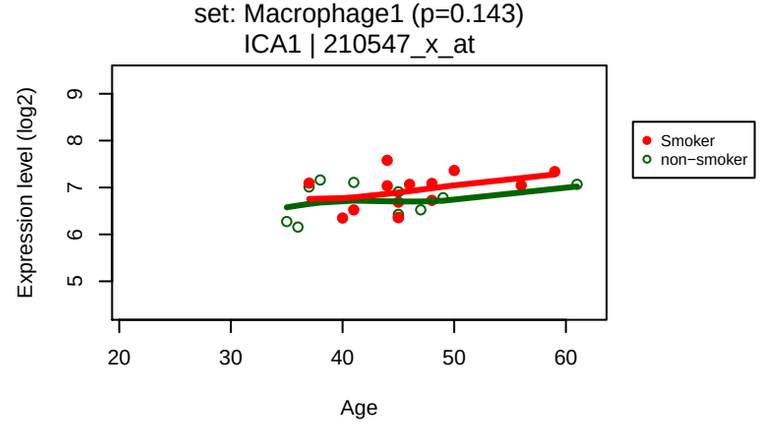

set: SmallAirway1 (p=0.5123)
ICA1 | 210547_x_at
Expression level (log2)
Age
Smoker
non−smoker

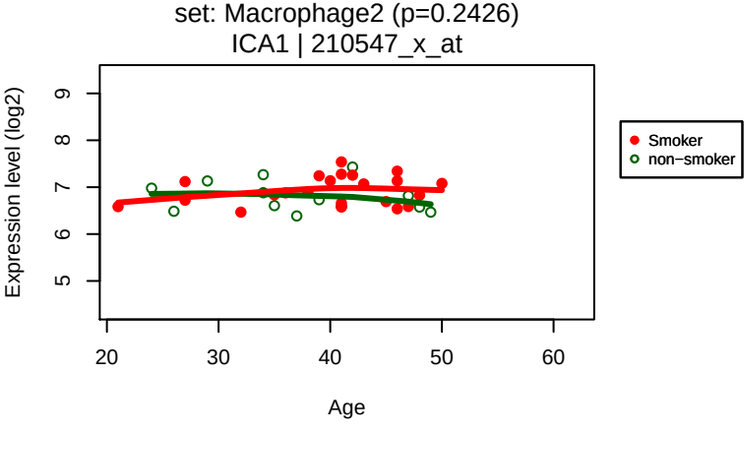

set: SmallAirway2 (p=0.7025)
ICA1 | 210547_x_at
Expression level (log2)
Age
Smoker
non−smoker

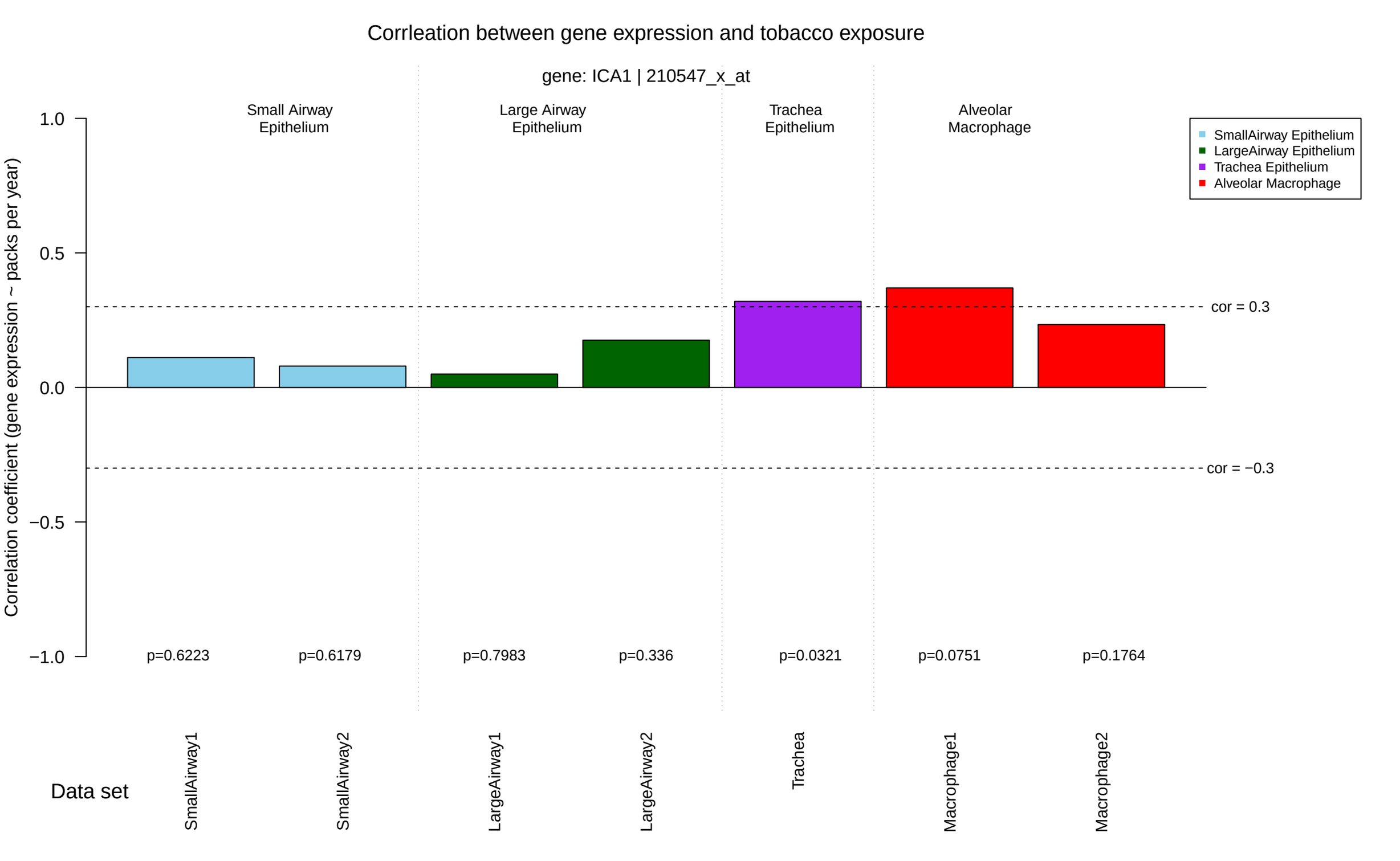

set: LargeAirway1 (p=0.7722)
ICA1 | 210547_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3276)
ICA1 | 210547_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0216)
ICA1 | 210547_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.143)
ICA1 | 210547_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2426)
ICA1 | 210547_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ICA1 | 210547_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6223
p=0.6179
p=0.7983
p=0.336
p=0.0321
p=0.0751
p=0.1764
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2